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Effectivity: An alternative to conceptions of gifted under- and over-achievement

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Abstract
This article introduces the concept of *effectivity,* which is the power to be effective or to achieve a certain effect to accomplish one's goals, in this case, desired school or life achievement. Giftedness would be better defined and developed in terms of effectivity than in terms of various concepts of ability. The concept of effectivity hinges on the idea that what determines, or what influences achievement, includes cognitive ability, but only as a small part of the total picture. Gifted "underachievement" as well as "over-achievement" disappear when one understands giftedness in terms of effectivity.

Keywords
Effectivity, achievement, underachievement, overachievement, ability, motivation, self-efficacy, resilience

John has a measured IQ of 130, placing him in the top 2% or so of the population of school children, but he is doing only slightly above average work in school. He has a 3.2 average in his school on a 4.0 scale, which places him only in the top 40% of his class. His school views him as an underachiever. His fraternal twin sister, Jeanette, has a measured IQ of 110, which places her in the top quarter of her class, but she has a 3.9 average in her school, which places her in the top 3% of her school. Although she has not been formally classified as such, she is viewed as an overachiever—or, as one of her teachers has put it, a

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"grind." John and Jeanette are hypothetical, but their respective situations are not, especially with regard to the hypothetical constructs of "underachievement" and "overachievement."

Two concepts in the literature on educational and school psychology are "underachiever" and "overachiever." Although there are various definitions of these concepts, they are usually defined such that an underachiever works at a level below their measured ability, and an overachiever works at a level above their measured ability (Karnes et al., 1961).

This article claims that both concepts—*underachiever* and *overachiever*--are flawed because the ability measures, on which the claims of underachievement or overachievement are based, are flawed. The flaw in the argument is that tests of abilities measure only a small part of what it takes to achieve successfully, in or outside school. They are not complete measures of overall potential or even of abilities, and they certainly are not complete measures of other characteristics of individuals that are associated with success. So labeling children based on narrowly conceived ability tests represents an injustice to the children, so-called "gifted" or not. The measures that the ability-test scores are compared to—school grades and achievement-test scores—have flaws of their own, but those flaws are not the main focus of this essay.

How one will look at gifted underachievement and overachievement will depend in part upon the model of giftedness one uses. Our essay is based on a conventional model that, at least in the United States, is overwhelmingly used in schooling (see discussions in Sternberg & Ambrose, 2021). We follow the T-ACCEL model of Sternberg (2024). Other similar models, we believe, such as the Munich Model of Heller et al. (2005), the actiotype model of Ziegler (2005; Ziegler & Stoeger, 2007), the three-ring model of Renzulli (1986; Renzulli and Reis, 2014), and the megamodel of Subotnik and her colleagues (2011, 2021) might well lead to conclusions similar to the ones we describe in this article. Our goal in this model is not to push any one particular model of giftedness over another, including our preferred one, but rather to point to challenges resulting from the use of a conventional model based on the work of Terman (1925) and other theorists and practitioners who have derived notions of under- and overachievement from IQ and its proxies. A number of broader, modern models could lead to conclusions similar or identical to ours, but their originators would have to be the ones to decide that.

This article will discuss problems in the way tests of intelligence and achievement are sometimes used in education. We wish to make clear, at the outset, that we believe both kinds of test can have positive, if limited uses. For example, intelligence tests such as the Wechsler Intelligence Scale for Children (WISC) in its various editions can be useful for providing a profile of a child's pattern of cognitive strengths and weaknesses at a given time in the child's life (Kaufman, 1979). Achievement tests can be useful for pointing out strengths and weaknesses in terms of children's achievement in various subjects, such as language, mathematics, science, history, and the like (McCormick, 2013). Our arguments below concern not the tests, in and of themselves, but rather their use for assigning the labels "underachiever" and "overachiever."

Concepts of under- and over-achievement

Our contention is that under- and over-achievement are largely illusionary constructs imposed on the educational world, which do not accurately characterize the psychology of children's performance, whether the children are gifted or otherwise. Regardless of whose "fault" one views under- or overachievement to be—the student, the family, the school, the society, or whatever--, one cannot really assign fault based on a construct that does not hold up psychologically or educationally.

We state one thing at the start: The idea of general intelligence (or *g,* also called "general mental ability"), as summarized by a single number, such as IQ (which also contains elements beyond *g*), is baked into much of the field of giftedness. That idea has been around since Terman (1916, 1925; Terman & Oden, 1959). Textbooks on giftedness (e.g., Rinn, 2020; Silverman, 2012) tend to fall back on IQ, even when considering giftedness broadly. But there are many other views of intelligence (see essays in Sternberg, 2020b), just as there are multiple and diverse views of giftedness. However, much of the concept of under- and over-achievement *assumes* the validity of IQ and related measures as indices of ability. If this assumption is inadequate, both the conceptualization and operationalization of under- and overachievement as they exist in much of the field must be questioned.

The inadequacy of indices based on measures of IQ and *g* for measuring ability

Indices of under- and over-achievement usually start with a difference score or derivative of difference scores between intelligence and some measure of achievement. A major problem with this conception is that intelligence is more than IQ or a unitary *g*.

The most eminent scholars and, arguably, the greatest minds in the history of the field of intelligence—including those who developed intelligence tests--recognized the inadequacy of IQ and general intelligence (*g*) as providing comprehensive or even adequate measures of intelligence (e.g., Baltes, 1987; Baltes et al., 1999; Binet & Simon, 1916; Cattell, 1971; Guilford, 1967, 1982; Hebb, 1949; Horn & Hofer, 1992; E. Hunt, 1976; J. McV. Hunt, 1961; Jensen & Munro, 1979; Luria, 1976; Minsky, 1988; Newell & Simon, 1972; Piaget, 1950, 1952; Schank & Morson, 1995; Spearman, 1923; Thomson, 1916, 1939; Thorndike & Stein, 1937; Thurstone, 1924; Vernon, 1955; Vygotsky, 1962/2012, 1978; Wechsler, 1943). This list includes most of the major intelligence theorists of past times.

One might think that the constructors of IQ tests would have been those who most stuck to them in their full conceptions of intelligence, but that inference turns out to be false. Spearman (1923) proposed qualitative and quantitative principles of cognition that went beyond *g*. Thomson (1939) argued that *g* is a statistical artifact. Binet and Simon (1916) presented a detailed theory of intelligence as judgment and what we today would call "metacognition," which went well beyond what their test tested. Thurstone (1924) discussed the importance of inhibition of instinctual responses to stimuli as key to intelligence (as did the evolutionary theorist, David Stenhouse, years later). Cattell (1971)

discussed the importance of the environment on intelligent responses. Guilford's (1982) theory had 120, then 150, then 180 factors, not all of which possibly could be measured by an IQ test. The list goes on. The most eminent theorists and testers of intelligence were not locked into IQ but believed that there is more to intelligence than IQ. Binet even planned to develop mental orthopedics to help children develop their intelligence, but he died before he was able to put his ideas into practice.

One might argue that this "Who's Who" of major intelligence theorists and researchers represents minds and knowledge of a past era. Perhaps those using intelligence tests and their proxies, which measure largely the same thing (Frey & Detterman, 2004; Kaufman et al., 2012; Koenig et al., 2008; Sackett et al., 2020) for estimating under- and over-achievement simply know something these distinguished theorists did not know in their day. But even the most widely accepted current theories of intelligence, Carroll's (1993) theory and CHC theory (McGrew, 2005, 2009; McGrew & Wendling, 2010; Schneider & McGrew, 2012) include a range of intellectual abilities that goes far beyond what IQ tests measure. Examples are communication ability (CM), oral production and fluency (OP), and knowledge of behavioral content (BC). Although these abilities and many others in the broader theories are characterized as subfactors of general intelligence (*g*), measurement of most of the abilities is not to be found in any of the major contemporary intelligence tests or their proxies.

Beyond the psychometrically based theories are also broader modern theories of multiple intelligences (Gardner, 2011), personal intellilgence, and emotional intelligence, which also expanded the range of thinking about intelligence (Mayer et al., 2004; Rivers et al., 2020). Others have explored multiple intelligences (Gardner, 2011; Kornhaber, 2020), successful intelligence (Sternberg, 2020), collective intelligence (Malone & Woolley, 2020), leadership intelligence (Boyatizis, 2020), cultural intelligence (Ang et al., 2020), practical intelligence (Hedlund, 2020; Sternberg et al., 2000), and intelligence as it applies in the real world (Ackerman, 2023; Kovacs & Conway, 2019), social intelligence (following Thorndike; Kihlstrom & Cantor, 2020), consumer and marketer intelligence (Sujan & Sujan, 2020), and mating intelligence (Geher et al., 2020).

It turns out that an astonishing number of the major (high-impact) theorists and searchers in the field of intelligence, both historical and contemporary, with scholarly citations in the millions, are being ignored by those who use IQ or *g* as a basis for estimating under- and over-achievement. On whose work is their use of IQ, *g,* or a proxy based, then? What renowned expert is behind the notion that IQ is an adequate measure, in itself, of a person's abilities for school achievement or anyplace else?

It turns out that the "expert" is the late Edwin Boring, who wrote a paper (Boring, 1923) on "Intelligence as the tests measure it." The paper should have met with skepticism. First, it was published in a popular magazine, *New Republic,* which was not exactly a scholarly journal but rather a popular magazine. Boring (1961) much later wrote a scholarly chapter on the same topic, based on his popular article. Usually, things go the other way around, with popular articles following scholarly ones. Second, Boring was not an expert on intelligence. He studied various things—sensation, perception, consciousness—but intelligence was not one of them.

If underachievement and overachievement were being studied only in the laboratory or were objects of theorization, the consequences would not be so great. But children are being labeled based on an operationalization that is seriously at odds with the thinking of an overwhelming number of the best minds in the field of intelligence, past and present. In an equation, if one term is invalid for the computation, the computation itself is invalid. In our view, it is irresponsible to take the difference between IQ and its proxies, on the one hand, and measures of achievement, on the other, and to declare the result a manifestation of "under-" or "overachievement."

Worse, sometimes the declarations are based on simple difference scores, which are notoriously unreliable. This fact was initially shown by two of the most eminent researchers in the field of psychometrics, Frederick Lord (Lord, 1958) and Lee Cronbach (Cronbach & Furby, 1970). Others may disagree, but to use such scores when they are known for unreliability seems questionable, at best.

Let us now turn to the issue of underachievement, which, as we have shown, is questionable in the first place.

The concept of underachievement

The greatest problem with the concept of *underachievement* is that psychologists have no measures of intelligence that are adequate to the concept of intelligence that are not also simultaneously measures of achievement (Cronbach, 1990; Sternberg et al., 2008). For example, tests such as the Stanford-Binet or Wechsler intelligence tests may include vocabulary, which is certainly achieved (Bus et al., 1995; Collins, 2014; Connor et al., 2005), arithmetic problem-solving and computation skills (Mattoon et al., 2015; Talib et a l., 2019), also achieved, and various kinds of abstract reasoning (Walker, 2015), which turn out to be achieved as well.

The intelligence tests may be compared with grades in school rather than achievement-test scores, but this procedure does exactly nothing to alleviate the problem. Assessments of school achievement measure achievement, plus many other things—motivation for scholastic work, social skills, ability to please teachers, willingness to please teachers, organization, self-efficacy, and match to who and what teachers believe "good students" should be. As discussed above, intelligence tests measure achievement plus somewhat narrow aspects of intelligence. Intelligence tests also measure, directly or indirectly, motivation to succeed on the tests, understanding of what is being asked, amount of Western schooling (Ceci, 1996), concentration, attention, comfort with the testing situation, stereotype threat (Steele & Aronson, 1995), and cultural match to the content of the test (Sternberg, 2004), among many other things. Two highly contaminated measures in a single equation do not provide a valid measure, no matter what the reliability of the difference measure.

Another important issue is that, for many children, achievement that is important to them is not achievement that is important to their school. A child who wishes to be a professional sculptor, dancer, orchestra or band musician, actor, numismatist, business executive, plumber, carpenter, graphic designer, or member of any of innumerable other professions might actually better spend their time developing their knowledge and skills

for their future profession than excelling in school. In music, one's window for getting into the field at a professional level is narrow. Practicing a musical instrument might be a better use of time than excelling in school subjects that will be irrelevant to career success. And then of course, those who seek asocial or anti-social occupations, such as drug dealers, may be achieving at high levels in ways that are not relevant to school success.

In many environments, adaptive success may be largely orthogonal to school success (e.g., Grigorenko et al., 2004) or actually opposed to school success (Sternberg et al., 2001). Oddly, the senior author of this article was an example of this principle. He spent much of his time and effort in high school studying psychology, which at the time was not taught or recognized as a valid subject of study in his high school. Yet, psychology became his later career. His time studying a subject that was utterly irrelevant to his school success, and took time away from his school success, proved to be time well-spent when he applied to college and later to graduate school.

At one time, it was thought—incorrectly—that tests of abstract reasoning or other kinds of nonverbal performance, such as visual analogies, series, or classifications—are culture-fair (Cattell, 1940, 1971; Cattell & Cattell, 1973) or even culture-free, because of the lack of verbal content. The nonverbal Army Beta test was designed for people from literally anywhere with any education or language skills who were willing to take the test (Yerkes, 1921). However, that presumption turned out to be false (Brandwein, 2011). Tests of so-called fluid abilities (ability to learn new concepts, reason abstractly, and solve novel problems), such as the problems that appear on nonverbal tests of intelligence, showed a greater Flynn effect (increase in scores) during the 20th century than did tests of crystallized (verbal) abilities (Flynn, 2012, 2016). These tests may show reduced cultural differences between immigrant and non-immigrant groups (e.g., Weiß, 2006). However, they are still based on abstract-reasoning skills (fluid intelligence) and showing whopping Flynn effects, indicating they are very susceptible to sociocultural influence. Formal deductive reasoning, also related to fluid ability (Guyote & Sternberg, 1981), is also a culturally learned skill. People apparently are not born to solve logic problems. Tests of intelligence alleged to be culture-free or culture-fair are neither culture free nor culture-fair, and are disadvantageous to many minoritized students, a point made by several cultural psychologists such as Michael Cole (Laboratory of Comparative Human Cognition, 1972), John Berry (1974), Alexander Luria (1976), Donna Ford (2004), Lisa Suzuki (Suzuki et al., 2017; Suzuki & Wilton, 2016), Fons van de Vijver (Thomas et al., 2015), and others.

It is further the case that tests of achievement often end up being measures, in part, of general mental abilities because those tests require much of the same knowledge and many of the same skills as achievement tests, resulting in all the tests correlating quite substantially with each other (see Frey & Detterman & 2004; Koenig et al., 2008; Sackett et al., 2020). Thus, the skills that were thought not to be susceptible to education, or at least Western education, are the *most* susceptible to development through such education. They certainly are not culture-free, culture-fair, or even, as it turns out, culture-reduced, as shown by the Flynn (1987) effect.

If one measures "underachievement" in terms of a difference between IQ, however determined, and scores on achievement tests, it is conceptually problematic because one is

looking at the difference between two measures, both of which, at some level, confound ability and achievement. The difference between IQ and achievement scores, statistically, is confounded because both measure a combination of abilities and achievement; any difference score thus is largely meaningless. However, laypersons and professionals alike, including psychologists, can be enticed to assign to test scores more meaning than they deserve because the scores appear to be quantitatively precise (Sternberg, 1985). Underlying such assignment of meaning is the illusion that, because numbers have been assigned, those numbers must be meaningful.

Not all measures that are purported to measure intelligence look like IQ tests. For example, there are measures of intelligence that are based on neuropsychological functioning (Haier, 2020), simple information-processing (e.g., tasks such as choice reaction time and visual inspection time--Nettelbeck et al., 2020), working memory (Conway & Kovacs, 2020), and other aspects of cognitive functioning (Sternberg, 1986). Our argument applies to these tests as well, not just to tests that superficially look like conventional tests of intelligence. But none of these tests has gained even narrowly based acceptance as an adequate measure of intelligence. In each case, it is arguable whether what any particular test is measuring even is intelligence rather than a correlate of it.

There are, of course, ways of identifying underachievers, and especially gifted underachievers, that bypass intelligence tests and their proxies (Whitmore, 1980). These ways include gaps between oral and written work, creativity that is not manifested in schoolwork, perfectionism that results in assignments not being turned in or turned in incompletely, low self-esteem, and more. The problem with these and other such indicators is that they do not uniquely identify underachievement. The phenomena could manifest themselves for many different reasons. Indeed, the studies that have been done make it difficult even to identify gifted underachievers (White et al., 2018).

On this view, it is unfair and psychologically and educationally unjust to identify someone as an "underachiever" when they may have developed skills other than those measured by conventional ability tests, as well as attitudes—such as creative and practical ones—that can be relevant for schoolwork but that are not identified by conventional ability tests. One may pejoratively label a child as an "underachiever" not because the child is truly an underachiever but rather because the child has not been properly diagnosed. The so-called "underachiever" may be achieving, just not in ways that are socioculturally sanctioned as constituting "achievement."

A fair question to ask, one perhaps far too infrequently asked, is why educators need the concept of *underachievement* at all, including *gifted underachievement.* The conceptualization seems weak in that it is not clear how one can know, given the current state of knowledge, what level of achievement one fairly can expect of an individual. It certainly is unclear how one can know whether that level of achievement is truly an underperformance based on some sense of what the true level of performance should be. As Gardner (2011), Sternberg (2021a), Renzulli (1986; Renzulli & Reis, 1997), and others have pointed out, current measures of abilities are narrow and conceptually quite limited. Moreover, it is not even clear that we have a viable notion of giftedness (Borland, 2005). Why would educators want to label a child as an "underachiever," with the pejorative connotations of this term, without a clear conceptual or operational basis underlying the

label? For example, young Brazilian street children may be acting in a gifted manner in running successful street businesses while failing in school (Ceci & Roazzi, 1994; Nuñes, 1994).

Consider further how badly the concept of *underachievement* can misinterpret a particular level of school performance. Take Tom and Mary, two hypothetical high school students whose school achievement, in each case, as measured by GPA, is substantially lower than what would have been expected from their IQs. Should the school be concerned? Perhaps not. John is proving to be a star not only of the school basketball team but also of state basketball competitions. His coach and others have told him that he has a reasonable probability of a career in professional basketball. Mary is the first violinist in her high school orchestra and a member of the All-State Orchestra for her state. She hopes to have a career, at best, as a violin soloist or, if not, as a professional violinist in a major orchestra.

Both John and Mary know they must spend much time practicing because deliberate practice is one of the most important ingredients in developing expertise (Ericsson & Pool, 2016). Both students are taking time from academic study to pursue an interest that could lead them to a successful career. Neither is "flunking out," but neither is performing at an expected academic level. From the perspective of their career aspirations, they likely are making the right decisions. They are achieving at high levels in the endeavors that matter most to them both at present and, potentially, in the future. Moreover, if one accepts elements of Gardner's (2011) theory of multiple intelligences (see also Sternberg, 2020), they are exercising their intelligences (John, bodily-kinesthetic; Mary, musical), not just engaging in haphazard or unintellectual activities.

A real case in point is a close relative of one of the coauthors of this paper. The relative was at a prestigious business school, and as he spent more and more time on a business start-up, his academic performance began to flag. He had done well in college but was not performing at the same level in business school. The coauthor was worried that he was mis-investing his time. Things seemed to get worse when a venture capital firm offered to fund his business, but only if he dropped out of business school and devoted full-time to the business. He dropped out of business school, a potentially costly move. However, the decision proved good, as he eventually leveraged the venture capital to become CEO of a multi-billion-dollar company he co-founded.

In each of these cases, a student whose academic performance might have signaled "underachievement" was rather achieving at a level that was not only high but directed toward their particular career and life aspirations. How, really, can educators even identify underachievers, should they exist, without considering what they, the students, are trying to achieve in their lives?

The Concept of Overachievement

The construct of *overachievement*—achieving at a level higher than one supposedly should, based on one's ability—is even less plausible than the construct of *underachievement.* If one is achieving at a level over that predicted by ability measures, the problem is with the measures, not with the individual. To denigrate the individual—perhaps

as a grind or a plod or simply as someone over-ambitious—does a tremendous disservice to the individual.

Suppose IQ, or the major construct underlying it, general mental ability (GMA), were equivalent to the complete set of abilities needed to achieve in school or elsewhere. In that case, one might entertain a concept such as overachievement. But does any educator really believe, or at least should any educator believe, that tests of IQ or their proxies are complete measures of abilities for success? In Gardner's (2011) theory, they do not measure or measure only minimally interpersonal intelligence, intrapersonal intelligence, naturalist intelligence, musical intelligence, and bodily-kinesthetic intelligence. Some of them do not even adequately measure spatial/visual intelligence. In Sternberg's (2021a) theory, they do not measure creative, practical, or wisdom-based intelligence and do not even fully measure analytical intelligence, which can differ in level depending upon the domain in which it is applied (see, e.g., Sternberg & Sternberg, 2017).

The theories of Gardner and Sternberg are broadly based, and many psychometricians prefer theories that do not extend as far as these two theories do, for example, preferring the Cattell-Horn-Carroll (CHC) theory of intelligence (e.g., McGrew, 2005, 2009). But if we look at this widely accepted psychometric theory, we find that even many of the abilities captured by that theory are also missing from conventional intelligence tests and their proxies. These abilities include, in many tests, reading ability (not measured by nonverbal or performance tests), writing ability, quantitative knowledge (unless there is an advanced mathematical section), various memory abilities (short-term and aspects of working memory), auditory processing, olfactory abilities, tactile abilities, psychomotor abilities, kinesthetic abilities, processing speed (except for strictly timed tests), decision speed/reaction time (except indirectly), and psychomotor speed. The tests mostly measure fluid and crystallized abilities, consistent with the earlier Cattell (1971; Cattell & Cattell, 1973) model of intelligence. In other words, even according to the most widely accepted psychometric theory of intelligence (at least in current times), it is unclear how one could infer "overachievement" because so many aspects of intelligence, by almost any contemporary model, are not measured by the existing tests of intelligence and their proxies.

Oddly, the concept of *overachievement* is almost directly contrary to the psychological thinking of the founders of the intelligence-testing movement, Alfred Binet and David Wechsler. Binet and Simon (1905, 1916) viewed intelligence as broad and malleable. Wechsler (1943) believed that intelligence comprises many non-intellective factors as well as the more intellective ones measured by conventional tests. More recently, Sternberg (2022; Sternberg et al., 2024), consistent with Wechsler, has proposed that intelligence comprises not only abilities but also a set of attitudes. Many people fail to act intelligently not because they lack particular abilities but rather because they lack certain attitudes, such as the importance of using their intellect when under the influence of runaway emotions and the importance of doing their best.

We add that there are psychometric and other theorists whose conception of intelligence is relatively narrow (Haier et al., 2023) and closely related to the work of Spearman (1904, 1927) begun more than a century ago. Oddly, Spearman's notion of intelligence was arguably broader than that of these theorists (Spearman, 1923). We view their notion of intelligence as out of touch with research across cultures and time periods (see

Sternberg, 2020c), which they largely ignore in favor of an extremely narrow base of theories and studies that presume the narrow perspective they bring to the field, and which has damaged the prospects so many children over the years (Ford, 1995; Gould, 1981; Lemann, 1999; Murdoch, 2007; Sternberg, Desmet, et al., 2021; Zenderland, 2001). This damaging view is that there is a single factor of intelligence, which operates in similar ways across cultures, and that people have relatively fixed levels that are unaffected by culture and life experience (Jensen, 1998). It is crucial that we challenge this narrow view and advocate for a broader perspective, as it will significantly impact children's prospects and is our responsibility as educators, researchers, and policymakers in the field of psychology and education.

If so many abilities, attitudes, and other aspects of the person that are viewed as part of intellectual abilities by various theorists in the field, including psychometric ones, are missing from tests used in making identifications, it is puzzling how overachievement could be measured by comparing scores on an IQ or a proxy, on the one hand, and measured achievement, on the other. By almost any standard, the measures of abilities used in schools are incomplete.

There is a particularly pernicious aspect to the construct of overachievement: the temptation to blame the child for failures of the educational system, whether in the testing the system or in the achievements that are considered worth monitoring. The range of achievements possible for children is enormous, and cultural studies show that what is valued for children's achievement varies from culture to culture (Sternberg, 2020b).

Replacing the concepts of under- and over-achievement: the concept of effectivity

What, then, does it take to succeed in schoolwork or life work? Are there reasons beyond "under-achievement" that might explain why some students of seemingly high ability—at least as measured by IQ tests and their proxies—might perform less well in school than expected? As we have seen, several factors can lead to the appearance of under-achievement or overachievement.

These factors, taken together, influence what we here call *effectivity,* which is defined as the power to be effective or achieve a certain effect (www.vocabulary.com), in this case, desired school or life achievement. The concept of effectivity is based on the notion that what determines, or at least influences, achievement includes cognitive ability, but only as a small part of the metaphorical equation. To refer to someone as under- or over-achieving based on a comparison between a score on a limited (ability) test and a score on another limited (achievement) test, or with school grades, presents a narrow view of the full range of knowledge, skills, attitudes, and other characteristics that underlie achievement, in school or anyplace else. What matters is not merely ability but also effectivity—one's overall potential to achieve effectively at particular desired levels of achievement. If one adopts effectivity as a notion, the notions of under- and overachievement go away. What are the elements of effectivity?

Effectivity is predicted by the interaction of personal, environmental, and task-related resources (see Figure 1). Personal resources include basic cognitive, metacognitive,

perceptual, learning, attentional, motivational, and personality resources. Task-related resources include achievement tests, everyday tasks, essays, projects and portfolios, and performances of various kinds. Environmental resources include levels of the environment, as described below.

Personal resources

Cognitive resources

These resources can be of different kinds.

Basic Cognitive Resources. General cognitive abilityis measured by conventional intelligence tests and tests of related abilities (Gottfredson, 1997; Herrnstein & Murray, 1994; Sackett et al., 2020; see essays in Sternberg, 2020a). As noted earlier, a challenge with conventional tests is that they measure limited abilities, even by the standards of what are now the most widely accepted conventional psychometric theories of intelligence (Carroll, 1993; McGrew, 2009). If one moves to broader theories of intelligence, the cognitive abilities they measure seem even more narrow (Gardner, 2011; Sternberg, 2021). For example, in Sternberg's theory of adaptive intelligence, certain school tasks might require one to think creatively—something not measured by conventional tests of abilities—or to think practically (commonsensically) or wisely—also skills not measured

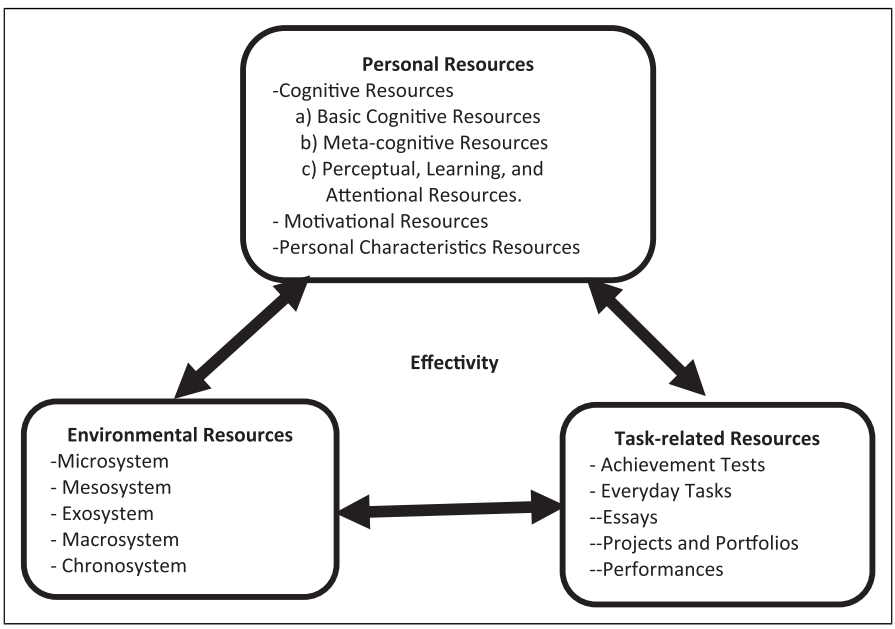

Figure 1. Effectivity elements.

by conventional tests of intelligence. So, one might seem to be underachieving with respect to IQ--which primarily reflects academic knowledge and abstract-analytical skills—when, in fact, one is not underachieving at all because one has not yet developed the creative, practical, or wisdom-based skills to succeed. Numerous examples of applications of creative and practical skills to school material can be found in Sternberg et al. (2008).

Metacognitive resources. Metacognitive resources are involved in evaluating the quality of work one is doing and one's progress, including both improvement and deterioration--in doing the work (Rhodes, 2019). Metacognitive resources are often viewed as executive processes in that they control other processes (Flavell, 1979). They also control when one chooses to use one's intelligence, creativity, or other cognitive skills and attitudes (Sternberg, 2021b; Sternberg, Glaveanu, et al., 2021). Used well, they allow anyone to improve their cognitive performance (Sternberg, 2010; Sternberg & Spear-Swerling, (1999). They include recognizing the existence of a problem, defining the problem, mentally representing the problem, allocating resources to the solution of the problem, monitoring problem-solving while it is ongoing, and evaluating the problem-solving after it is completed (Sternberg et al., 2008). They help one transition from being a novice to being an expert (Charness et al., 2006; Sternberg & Grigorenko, 2003). Moreover, they help one recognize when fast, automatic processing is preferable and when one needs to use more thoughtful and reflective information processing (Kahneman, 2011).

Conventional intelligence tests and their proxies measure metacognition, but only indirectly, for example, in how one allocates one's time in taking the test, which has become less of an issue in recent times, when items are administered one item at a time, rather than all together so that one had to allocate time across test items.

Teacher-made tests, however, often require a lot of metacognitive skills. One has a class period to answer a set of items, and one has to decide how to divide one's time across test items; how much time to spend on any one item, perhaps a problem, before deciding that one cannot solve it or that one has to write down an answer of which one is uncertain; what kinds of answers the teacher or other grader expects; and whether one's interpretation of a problem even is correct. Homework assignments require even more metacognitive skills, requiring one to balance off when they are due, how much they count, how much work to put into each one, what kinds of answers are expected, and so forth. One could be adept at the cognitive work required to solve problems without necessarily being adept at the metacognitive work of planning one's work, monitoring that work, and evaluating the work before handing in a given assignment.

Furthermore, studies have shown the importance of self-regulation (Baum et al., 1995; Neumeister & Hébert, 2003; Nota et al., 2004; Steenbergen-Hu et al., 2020) and self-monitoring (Siegle, 2022) in predicting students' performance. These abilities help students evaluate their work and ascertain their strengths and weaknesses. Interventions focused on enhancing students' self-regulation are often effective (Baum et al., 1995; Obergriesser & Stoeger, 2015; Siegle, 2022).

Perceptual, learning, attentional, and other cognitive resources. Almost all cognitive tasks require some level of sustained and carefully allocated attention. They require perceptual abilities to encode the problem, learning to figure out what kind of problem one is facing and how it might be solved. They require attention to learning so that students gain knowledge. Some individuals, however, have specific perceptual, learning, or attentional disabilities, even if they are gifted (Sternberg & Desmet, 2023).

Disabilities for which there are no proper accommodations further reduce achievement below that which might otherwise have been expected. Typical accommodations are extra time and a quiet room to take tests. However, there are many different kinds of cognitive disabilities, and the appropriate accommodations will vary. For example, students lacking mental well-being (Kurtz & Swenson, 1951), living with ADHD (Barry et al., 2002; Hinshaw, 1992), or having test anxiety (Ford, 1995) require different interventions. A student could have a high IQ, reflecting high cognitive abilities. However, if they cannot make out the words (reading disability / dyslexia) or numbers (mathematical disability/dyscalculia), they may find that they cannot perform at the level of which they are capable.

The issue is whether having a disability is a cause of underachievement or, rather, whether the cause is a school's or other educational institution's failure to make appropriate accommodations for the disability. A disability lowers effectivity if appropriate accommodations are not made. The word "underachiever" has the connotation that, somehow, the underachiever is responsible—that they are not working up to capacity. In the case of disabilities, if there is any meaningful form of underachievement, it is perhaps the school that is underachieving, in that the school should know both that accommodations need to be made and what kinds of accommodations to make.

Motivational resources

There are different kinds of motivations. Each person is motivated by different incentives (Neumeister & Hébert, 2003). People with higher scores on achievement tests tend to have a stronger inclination toward pursuing higher education and clearer strategies to achieve such education (Kuh et al., 2006). Encouragement from parents and teachers plays a role in what motivates children (Kuh et al., 2006). When children are encouraged to get good grades at school, they may become motivated by higher school achievements (Kuh et al., 2006). These students are also more worried about the risk of not obtaining the level of education they aspire to. They recognize the connection between education and their future lives, valuing it beyond just its job-related benefits (Kuh et al., 2006). Typically, by the upper grades, these children have well-defined career goals that are also realistic (Kurtz & Swenson, 1951).

There are many different theories of motivation (Elliot et al., 2017), but the ultimate role of motivation in effectivity is simple and direct: People, on average, achieve at lower levels on tasks they are not very motivated to do (Mega et al., 2014).

Personal characteristics resources

Some students have personal characteristics that help them succeed more than others in solving problems. They try to overcome difficulties and excel in their performance. However, some other students struggle due to personal characteristics that can hinder their academic progress. For instance, students with high levels of resilience generally have higher levels of progress in their studies (Backmann et al., 2019; Eells, 2017). Luthans (2002, p. 702) defined *resiliency* as "the positive psychological capacity to rebound, to 'bounce back' from adversity, uncertainty, conflict, failure or even positive change, progress and increased responsibility." Therefore, high-resilience students view challenges as opportunities for growth and learning new things rather than obstacles. The resilience mindset enables them to persist in adversity, maintain a positive outlook (Riley & Masten. 2005), and develop effective problem-solving strategies (Fan et al., 2021).

Other personal characteristics such as self-efficacy (Çikrıkci, 2017) and conscientiousness (Imhof & Spaeth-Hilbert, 2013; Kertechian, 2018; Rieger et al., 2022) also play a role in the achievements of students. Students with lower self-efficacy may doubt their abilities and feel less capable of achieving academic success (Doménech-Betoret et al., 2017). This lack of confidence can lead to increased anxiety, procrastination, and avoidance of complex tasks (Gadosey et al., 2021). Moreover, self-efficacy is closely linked to students' psychological well-being (Tang & Zhu, 2024), highlighting the broader implications of this personal characteristic. On the other hand, conscientious students are typically organized, responsible, and diligent. They are more likely to plan their study schedules, set realistic goals, and follow through with their commitments (Sorić et al., 2017). This level of discipline and attention to detail often translates into better academic performance and higher grades (Duckworth & Seligman, 2005).

Environmental resources

Environmental factors play a key role in the manifestations of students' talents (Ford, 2004; Matthews & McBee, 2007; Phillipson, 2008; Siegle, 2022). As seen in Figure 2, environmental factors can be considered at different levels of the environment (based on Bronfenbrenner, 2013).

Microsystem

The category is related to the factors in the immediate environment that directly interact with the student. For example, studies show that teachers, parents, and peers impact students' performance (e.g., Barton, 1997; Behrens & Vernon, 1978; Emerick, 1992; Kurtz & Swenson, 1951). Kurtz and Swenson (1951) proposed that home conditions include parents' interests in helping with their child's school activities and working with their child. Students with higher achievement tend to have better relationships and mutual respect with their parents. Kurtz and Swenson also argued that different children can perceive the same home and parent differently. Other scholars also support the notion of parents' effects on their children's academic success (Barton, 1997; Emerick, 1992). The

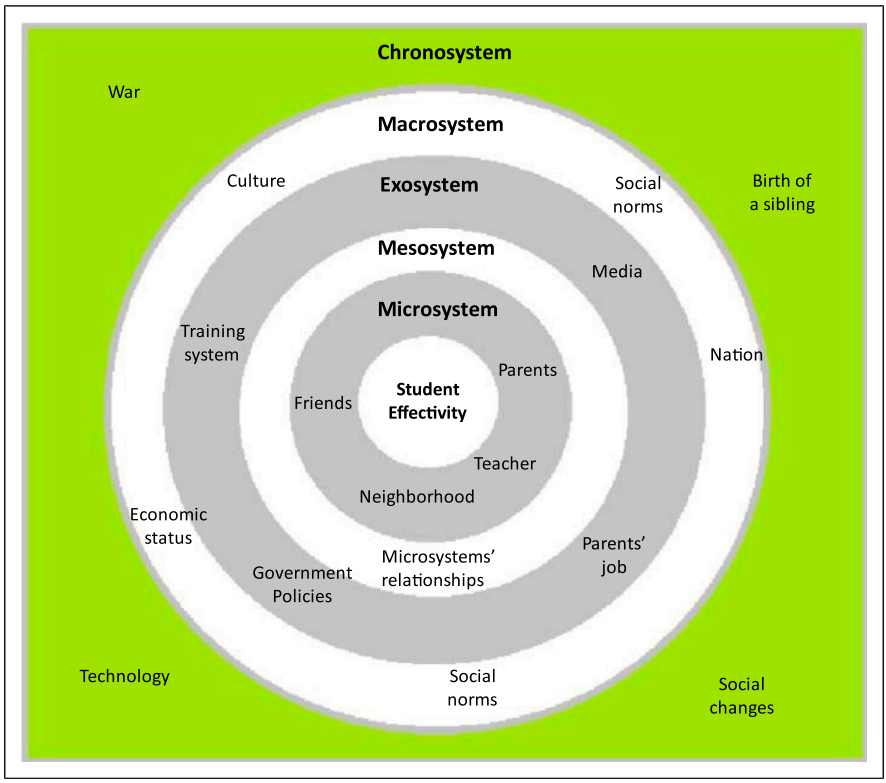

Figure 2. Environmental factors (based on Bronfenbrenner, 2013).

quality of peer relationships is also important for student achievement (Karnes et al., 1961). In addition, students with higher achievement tend to have friends concerned about school achievement (Kurtz & Swenson, 1951). Furthermore, the relationship between student and teacher also affects the student's performance (Postlethwaite & Haggarty, 2002). In all these examples, children's attitudes toward the school are as important as how children perceive their school and teachers (Behrens & Vernon, 1978).

Mesosystem

The mesosystem level indicates the relationship between different microsystems in a student's life, such as family, school, and peers. Optimal and positive interconnections between these systems are crucial for the student's academic success. For example, how well can parents communicate with teachers? Do the parents and school have the same expectations for the student? Do the parents accept their children's friends, and why or why not? Do these environments' goals align? When different microsystems' goals and expectations are aligned, students feel less conflicted and can focus on their achievements.

For example, Benner et al. (2021) found that high teacher expectations have more positive effects on students' academic achievements when parents also have high expectations for the students.

Exosystem

The exosystem level refers to an environmental system that indirectly affects children's lives without children being directly part of it. For example, parents' jobs and employment are predictors of children's performance at school (Neumeister & Hébert, 2003), but children do not work for their parents' employers. Children of overly busy parents who cannot provide sufficient attention often have lower achievement (Obomanu & Adaramola, 2011). Moreover, parents' education is another key factor affecting children's achievement (Neumeister & Hébert, 2003). The government also can affect children's achievement. Some governments, at any level, have high expectations for students and tests to ensure that the students meet these expectations. Other governments have no particular expectations or may have expectations that they do not reinforce or even enforce. Moreover, the expectations may or may not reinforce children's learning of what, ideally, they need to learn. For example, in autocratic regimes, much of the instruction may be oriented toward obedience to the regime and toward disdain for, or even hatred of, people living under other regimes.

Macrosystem

The macrosystem level refers to norms on a large scale, such as cultural norms, social norms, and broadly based classroom norms (Postlethwaite & Haggarty, 2002). Unfortunately, stereotypes, including negative ones, may be conveyed by the macrosystem (Jones & Myhill, 2004). Social and cultural expectations and norms help to determine children's perceptions (Ford, 1992). Discrimination, prejudice, low socioeconomic status, and cultural values differing from mainstream culture can lead to lower achievement scores (Sirin, 2005). For example, when the values and norms of Black members' culture are inconsistent with the values of the majority, which also is the majority culture that is dominant in most school systems, students often show lower levels of achievement (Ford, 1992).

These results also align with the stereotype threat concept (Croizet et al., 2001; Hartley & Sutton, 2013; Steele & Aronson, 1995) and how it can harm school achievement. This construct suggests that people usually react in a way that the stereotype suggests that they will act. This phenomenon applies to different races, ethnicities, and genders (Jones & Myhill, 2004; Steele, 1997).

Chronosystem

This category refers to the major changes that might occur in the environment during the child's life, such as war, political crisis, the birth of a new sibling, and the like. These factors can either advantage children (e.g., an increase in social class) or disadvantage

them (e.g., a war). Moreover, the secular time in which one lives can seriously affect what is possible for one's life. People in preliterate societies who might have become great writers later on would never have gotten a chance to find out, nor would posterity find out.

Task-related resources

Researchers use different kinds of tools to evaluate alleged over-achievement and under-achievement. However, all of the tools focus on virtually the same thing and consider academic achievements or cognitive abilities, such as those measured by IQ and achievement tests (for example, the WISC--Abelman, 2007; high school achievement tests--Abu-Hamour & Al-Hmouz, 2013; see also Reis et al., 2005). The definition of over- and under-achievement heavily depends on the current tools used to evaluate students. Therefore, using only limited tools, such as IQ and school achievement tests for evaluating students' progress, has some limitations:

1. Evaluation of over/underachievement is limited to those tools. However, as noted earlier, children have a variety of talents (Gardner, 2011; White et al., 2018).
2. Academic achievement predicts only weakly people's success in the job market. In other words, the skills students learn in school are often quite different from those needed to solve real-world problems (McGunagle & Zizka, 2020; Sapaat, 2011).
3. IQ and achievement tests are ability-based measures. Studies showed that another component of intelligence is attitude, which should be considered when interpreting the results of the tests (Sternberg, 2022; Sternberg et al., 2024). Studies showed that the attitude of children toward school and grades predicts their grades (Matthews & McBee, 2007). This finding, in turn, supports the idea that students might have the ability to succeed but not deploy it. Therefore, it is important to consider this component when evaluating students' performance.
4. Students might use most of their resources in activities apart from school. Having meaning in life is important for many of them—their sense of direction may differ from the direction the school is leading them in. Some students may believe that grades do not give meaning to their lives. Kurtz and Swenson (1951) argued that not all students with lower academic achievements cannot read or study well. However, many of these students are more interested in reading and studying things that schooling does not require. They may devote their time to other topics, and school programs should be flexible enough to cover different kinds of interests. Therefore, if schools become broader in education and cover more topics, differences in achievement might decrease.

Interventions for directing effectivity toward positive activities

Many factors affect students' performance, such as an appropriate curriculum that motivates students (Baum et al., 1995), having a growth mindset (Jiang et al., 2023), and a sense of purpose (Paunesku et al., 2015), flexible and varied topics available to students (Kurtz & Swenson, 1951), and having good mental health (Barry et al., 2002; Kurtz &

Swenson, 1951). These factors can be categorized into three categories: Person, Environment, and Task. Therefore, an effective intervention should consider all these factors. However, above all, we need to develop future adults who will seek a common good and seek to make positive changes in the world. When seeking the common good is considered the goal of education, the three factors of effectiveness will be oriented toward directing children in the kind of positive direction that is needed for the world.

Conclusion

Gorard and Smith (2004) have stated that not all underachievers are the same. Balduf (2009) emphasized that underachievement is a sign that education does not work for all students. Balduf's study showed that over/under-achievement concepts are inadequate and have limitations. Instead, focusing on effectiveness, a combination of Person, Environment, and Task-related factors, is better for students and society because it covers more aspects of students' performance. The concepts of Over/underachievement are derived from the current tools used, while the variety of talents and values among people is greater than what those tools can show. Focusing on the results of achievement tests misleads scholars. What is important in the world is not merely the grades of students but, above all, their overall contributions in life, based on their unique talents, the tasks they face in the environment they are living in, and, in the end, the positive change they can make to the world. In the end, what matters most is not particular abilities or achievements but rather effectivity—one's ability to affect the world in ways that reflect one's (hopefully positive) purposes and goals.

We present effectivity as a construct that considers how effectively a student (or, in principle, anyone else) can do the work they need to do. Because students need to do different work at different grade levels, in different specializations, in different subjects within those specializations, in different kinds of learning and performance activities, and different sociocultural settings, we doubt that there is any single measure, or even kind of measure of effectivity, that will work across the board, in schools or elsewhere. But, the field of intelligence measurement, historically, has been dominated by measuring first and then coming up with theories later, dating back to Spearman (1904) and even earlier to Galton (1883) and James McKeen Cattell (1890). When children's educational futures and, likely, occupational futures are at stake, is a seriously flawed measurement better than none? We think not. The use of the concepts of under- and over-achievement has gone contrary to the theories and research of many, if not most, of the most eminent psychologists, past and present, in the field of intelligence. It has been operationalized in a near theoretical vacuum.

An alternative approach would be to admit that measures of ability and achievement are confounded with each other and with other factors and that instead of relying on such measures, we in the West should take a more Japanese-based approach to education (Stevenson & Stigler, 1994) that encourages each student to work hard to achieve at the maximum level possible of what we are calling effectivity. We might find that some students relegated to the "underachiever" category could succeed, and we would eliminate the "overachiever" category. It is not a novel idea: Just encourage students to put

forth their best effort and thereby show that, if they do their best, and if teachers support them, they almost always can find a way to succeed. It has worked well in Japan, Finland (Colagrassi, 2018), and other countries whose student educational levels exceed those of the United States and much of Europe. Perhaps it can work where you are!

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